IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS FOR ENGINEERS IIA

Tuesday, 6 January 2026, 10:00 AM

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer ALL questions.

Question ONE is worth 20 marks. Question TWO is worth 30 marks.

NO CALCULATORS ALLOWED.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W.T. Pike

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ELEC50011A - MATHEMATICS 2A-2026-QUESTIONS

1. a) Given the function $u(x,y) = -e^{-y}\sin x - e^{-x}\cos y$,

(i) show that $u(x,y)$ satisfies Laplace's equation and integrate the Cauchy-Riemann equations to find its harmonic conjugate $v(x,y)$; [5]

(ii) hence obtain $f(z) = u(x,y) + iv(x,y)$ where f is an analytic function of $z = x + iy$, simplifying as much as possible. [5]

b) Let a mapping from the $z-$plane to the $w-$ plane be given by

$$w(z) = \frac{z}{i - z}.$$

Show that a family of circles C_1 in the $z-$plane, with centre $z = i$ and radius r, maps to a family of circles C_2 in the $w-$plane, stating their centre and radius.

Explain carefully what happens to points inside and points outside C_1 under the mapping. Finally, state the images under the map of the vertical axis, and the line $y = 1$. [10]

2. a) Given the real integral

$$I = \int_0^{2\pi} \frac{d\theta}{(5+4\sin\theta)^2},$$

(i) Taking complex z, where $|z| = 1$, show that we can rewrite I as a contour integral

$$I = \frac{1}{a} i \oint_C \frac{z\,dz}{(z+bi)^2\left(z+\frac{1}{b}i\right)^2},$$

giving the closed contour C, and integers a, b. [5]

(ii) Using Cauchy's residue theorem, or otherwise, calculate I. [5]

Recall that the residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z\to a}\left\{\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$

b) Prove that real, symmetric matrices have real eigenvalues. [6]

c) Obtain the orthogonal diagonalization of the symmetric matrix

$$A = \begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}. \quad [6]$$

d) Given

$$A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \end{pmatrix},$$

its Singular Value Decomposition (SVD) is $A = U\Sigma V^T$. Obtain U, V and Σ, stating which bases of fundamental subspaces are involved, and use these to write A as a sum of rank-1 matrices. [8]